# The odd-indexed terms of a nested square are even

A proof of an observation in OEIS A392210

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**Abstract**

The sequence A392210 of the On-Line Encyclopedia of Integer Sequences (OEIS), contributed by Paul D. Hanna, is defined by an "infinitely nested square"

$$x + \mathrm{sq}\Big(\frac{x}{1-x} + \mathrm{sq}\Big(\frac{x}{1-2x} + \mathrm{sq}\Big(\frac{x}{1-3x} + \cdots\Big)\Big)\Big), \qquad \mathrm{sq}(y) = y^2.$$

Its terms are $1, 1, 4, 18, 96, 579, \ldots$, and the entry observes that the terms $a(2n-1)$ with $n > 1$ appear to be even. We first make the nested expression precise: cutting it off after $d$ levels gives the correct coefficients up to degree $d+1$, and the resulting series is the unique solution of the simpler equation $A(x) = x + A\big(x/(1-x)\big)^2$. We then prove the observation. The proof is short: once all coefficients are reduced modulo 2, the square of any power series contains only even powers of $x$. The same idea also gives a recursive rule for the parity of the even-indexed terms. The main results have been checked by computer and proved formally in the Lean proof assistant.

**How this paper was produced.** This paper reports the result of an autonomous AI research effort. ChatGPT (OpenAI), using the GPT-6 model with Pro reasoning and working from prompts by Dev D. Goyal, found the proof, wrote its Lean 4 formalization, and wrote a first draft of the paper. Dev D. Goyal chose the problems and asked for checks and revisions, but did not supply proof ideas. Claude Opus 5.5, used with Claude Code, then checked the work independently: it rebuilt the formal proofs from source and confirmed that they rely on no unproved assumptions, recomputed the sequence from its definition and tested every claim against the published data, and read every step of the proofs. It also rewrote this paper for clarity. Dev D. Goyal has verified the results and is responsible for them. The paper has not been peer reviewed. The section "About this paper" at the end gives the details.

# 1 Introduction

## 1.1 The sequence

The *On-Line Encyclopedia of Integer Sequences* (OEIS) [2] is a large public database of integer sequences, used by mathematicians and scientists to identify sequences and to record what is known about them. Many entries also record observed patterns that the contributor could not prove.

This paper is about entry A392210 [1], contributed by Paul D. Hanna in February 2026. The entry reads, in part:

```
G.f.:  x + sq(x/(1-x) + sq(x/(1-2*x) + sq(x/(1-3*x) + sq(x/(1-4*x) +
...))))  where sq(x) = x^2, an infinitely nested square.
```

> This sequence equals the self-convolution (shifted right) of the binomial transform of this sequence.
>
> It appears that a(2*n-1) is even for n > 1.
>
> (1) A(x) = x + A(x/(1-x))^2.
>
> (5) A(x/(1-n*x)) = x/(1-n*x) + A( x/(1-(n+1)*x) )^2 for all n.
>
> — OEIS A392210, Paul D. Hanna, February 2026

Here "G.f." stands for *generating function*: the terms $a(1), a(2), a(3), \ldots$ of the sequence are stored as the coefficients of the power series $A(x) = a(1)\,x + a(2)\,x^2 + a(3)\,x^3 + \cdots$. The expression $\frac{x}{1-x}$ stands for the power series $x + x^2 + x^3 + \cdots$, and more generally $\frac{x}{1-cx} = x + cx^2 + c^2x^3 + \cdots$. Section 2 explains these notions.

The first line defines the series by an *infinitely nested* expression: a square inside a square inside a square, without end. It is not obvious what such an expression means. Section 3 gives a precise meaning, by cutting the nesting off after finitely many levels, and shows that the result is the same as the series defined by formula (1) of the entry,

$$A(x) = x + A\Big(\frac{x}{1-x}\Big)^2.$$

Here $A(\frac{x}{1-x})$ means that the series $\frac{x}{1-x}$ is substituted for $x$ in $A$, and the result is then squared in the ordinary way. We take formula (1) as the definition.

The second line of the quotation says the same thing as formula (1), in words. Substituting $\frac{x}{1-x}$ into $A$ gives the series whose coefficients are

$$b(n) = \sum_{k=1}^{n} \binom{n-1}{k-1} a(k),$$

which is called the *binomial transform* of the sequence. Squaring a series combines its coefficients into the sums $\sum_k b(k)\,b(n-k)$, called the *self-convolution*. So formula (1) says that $a(1) = 1$ and, for $n > 1$,

$$a(n) = \sum_{k=1}^{n-1} b(k)\,b(n-k),$$

which lets one compute the terms one at a time. The first terms, starting at $n = 1$, are 1, 1, 4, 18, 96, 579, 3834, 27404, 209052, 1688389, ...

### 1.2 The question

The entry observes that $a(2n-1)$ appears to be even for every $n > 1$, that is, that $a(3), a(5), a(7), \ldots$ are all even. The table shows the parity of the first terms (1 means odd, 0 means even). The odd indices $n \geq 3$ are printed in bold.

| $n$ | 1 | 2 | **3** | 4 | **5** | 6 | **7** | 8 | **9** | 10 | **11** | 12 | **13** | 14 | **15** | 16 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $a(n)$ mod 2 | 1 | 1 | **0** | 0 | **0** | 1 | **0** | 0 | **0** | 1 | **0** | 1 | **0** | 1 | **0** | 1 |
| $n$ | **17** | 18 | **19** | 20 | **21** | 22 | **23** | 24 | **25** | 26 | **27** | 28 | **29** | 30 | **31** | 32 |
| $a(n)$ mod 2 | **0** | 1 | **0** | 1 | **0** | 1 | **0** | 0 | **0** | 1 | **0** | 1 | **0** | 1 | **0** | 1 |
| $n$ | **33** | 34 | **35** | 36 | **37** | 38 | **39** | 40 | **41** | 42 | **43** | 44 | **45** | 46 | **47** | 48 |
| $a(n)$ mod 2 | **0** | 1 | **0** | 1 | **0** | 1 | **0** | 0 | **0** | 1 | **0** | 1 | **0** | 1 | **0** | 0 |

### 1.3 What this paper proves

**Main results.** *(1) The nested expression of the title has a precise meaning, and it defines the same series as formula (1) (Theorem 3.1). (2) The coefficient $a(2n-1)$ is even for every $n > 1$ (Theorem 3.2).*

The reason for (2) is simple. Modulo 2, the square of a power series contains only even powers of $x$. So, by formula (1), modulo 2 the only odd power of $x$ in $A(x)$ is the single term $x$.

### 1.4 What was known

The entry lists eight equivalent formulas for $A$, among them formula (1), formula (5), and the nested expression of the title, and states the evenness of $a(2n-1)$ as an observation ("It appears that"). The observation follows quickly from formula (1), and it may well have been known to the contributor; we are not aware of a written proof. We include it for completeness, together with a precise treatment of the nested expression.

### 1.5 Why it is interesting

Infinitely nested expressions, such as nested squares, nested square roots, or continued fractions, are a common way of describing power series in the OEIS. They raise a natural question: what does the expression mean? For power series there is a simple and general answer, which we explain here: each additional level of nesting fixes one more coefficient, so the coefficients of the truncated expressions eventually stop changing. The parity argument, in turn, is a basic tool that is used throughout this collection of papers: modulo 2, squaring a series doubles all its exponents.

### 1.6 How to read this paper

Section 2 explains the few notions we need: power series, substitution, the fractions $\frac{x}{1-cx}$, and remainders. Section 3 states the results precisely, and Section 4 proves them. Section 5 describes how the results were checked, by computer and by a formal proof in Lean. Section 6 discusses the scope of the result, and the parity of the even-indexed terms.

## 2 Background

This section collects the definitions and facts used later. Readers who know formal power series and congruences can skip it and return when needed.

### 2.1 Formal power series

A *formal power series* is an expression

$$U(x) = u_0 + u_1 x + u_2 x^2 + u_3 x^3 + \cdots,$$

determined by its sequence of coefficients $u_0, u_1, u_2, \ldots$. No number is ever substituted for $x$, so questions of convergence do not arise. We write $[x^n]\,U = u_n$ for the coefficient of $x^n$ in $U$; two series are equal when all their coefficients are equal.

Series are added coefficient by coefficient and multiplied like polynomials:

$$[x^n]\,(UV) = u_0 v_n + u_1 v_{n-1} + \cdots + u_n v_0.$$

Each coefficient of a product is a finite sum. For example, $(1-x)(1+x+x^2+x^3+\cdots) = 1$, so $1+x+x^2+\cdots$ is the *multiplicative inverse* of $1-x$, written $\frac{1}{1-x}$. More generally, every series $D$ with constant term 1 has a unique multiplicative inverse $\frac{1}{D}$, whose coefficients are determined one at a time by $D \cdot \frac{1}{D} = 1$ using only addition, subtraction and multiplication. Every fraction in this paper has a denominator of this kind, so it denotes a power series.

We write $U = O(x^d)$ when all coefficients of $U$ below degree $d$ are zero. So $U - V = O(x^d)$ means that $U$ and $V$ agree up to degree $d-1$. A series of the form $x + u_2 x^2 + u_3 x^3 + \cdots$, with constant term 0 and coefficient of $x$ equal to 1, is called *normalized*.

### 2.2 Substituting one series into another

If $V(x)$ has constant term 0, we can substitute $V$ for $x$ in $U$:

$$U(V(x)) = u_0 + u_1 V(x) + u_2 V(x)^2 + u_3 V(x)^3 + \cdots .$$

This is called *composition*, also written $U \circ V$. It makes sense because $V(x)^k$ has no terms below $x^k$, so only the finitely many terms with $k \le n$ affect the coefficient of $x^n$. For example, if $U = x + x^2$ and $V = x^2$, then $U(V(x)) = x^2 + x^4$. Composition is associative: $U \circ (V \circ W) = (U \circ V) \circ W$.

Substituting a series into itself repeatedly gives its *iterates*: $U^{\circ 2}(x) = U(U(x))$, $U^{\circ 3}(x) = U(U(U(x)))$, and so on, with $U^{\circ 0}(x) = x$ and $U^{\circ 1} = U$. They must not be confused with *powers* $U^2 = U \cdot U$, $U^3 = U \cdot U \cdot U$. (The OEIS sometimes writes $A^k(x)$ for the $k$-th iterate. In this paper the small circle always means an iterate, and a plain exponent always means a power.)

### 2.3 The fractions $x/(1-cx)$

For a number $c$ we write

$$M_c(x) = \frac{x}{1-cx} = x + cx^2 + c^2 x^3 + c^3 x^4 + \cdots .$$

In particular $M_0(x) = x$. These series behave very simply under substitution:

$$M_c(M_d(x)) = M_{c+d}(x). \tag{1}$$

To see this, start from $M_c(y)\,(1-cy) = y$, which is the definition of $M_c$, and substitute $y = M_d(x)$. Since $M_d(x)\,(1-dx) = x$, multiplying both sides by $1-dx$ gives $M_c(M_d(x))\,(1-dx-cx) = x$, which says exactly that $M_c(M_d(x)) = \frac{x}{1-(c+d)x}$. (Here we used that substituting into a product is the same as multiplying the substituted factors.) For example, substituting $\frac{x}{1-x}$ into itself gives $\frac{x}{1-2x}$.

Substituting $M_1$ into a series is the binomial transform mentioned in the introduction: since $M_1(x)^k = x^k(1-x)^{-k} = \sum_{n \ge k} \binom{n-1}{k-1} x^n$, the coefficient of $x^n$ in $A(M_1(x))$ is $\sum_k \binom{n-1}{k-1} a(k)$.

### 2.4 Remainders and congruences

For an integer $m \geq 2$ we write $a \equiv b \pmod{m}$ ("$a$ is congruent to $b$ modulo $m$") when $a - b$ is divisible by $m$, that is, when $a$ and $b$ leave the same remainder after division by $m$. For example, $17 \equiv 5 \pmod{4}$, since both leave remainder 1, and $-3 \equiv 1 \pmod{4}$. The remainder of $a$ on division by $m$ is called its *residue* modulo $m$; it is one of $0, 1, \ldots, m-1$. Saying that $a$ is odd is the same as saying $a \equiv 1 \pmod{2}$.

Addition and multiplication are compatible with congruences, so the residues $0, 1, \ldots, m-1$, added and multiplied modulo $m$, form a ring, written $\mathbb{Z}/m\mathbb{Z}$. When $m = p$ is a prime we also write $\mathbb{F}_p$. In $\mathbb{F}_2 = \{0, 1\}$ we have $1 + 1 = 0$, so addition and subtraction coincide.

For power series with integer coefficients, $U \equiv V \pmod{m}$ means that *every* coefficient of $U - V$ is divisible by $m$. Equivalently, the two series become equal when every coefficient is replaced by its residue; we call this *reducing* the series modulo $m$. The key fact is the following.

*Reduction modulo $m$ is compatible with all the operations above.* To reduce a sum, a product, a substitution, or an inverse, we may reduce the ingredients first and then compute in $\mathbb{Z}/m\mathbb{Z}$. This holds because every coefficient of the result is obtained from finitely many coefficients of the ingredients by adding, subtracting and multiplying, and these operations are compatible with remainders.

**Rings.** Several lemmas below hold over every *commutative ring*, that is, every set with an addition, subtraction and multiplication satisfying the usual rules of arithmetic (for example $\mathbb{Z}$, $\mathbb{Q}$ and $\mathbb{Z}/m\mathbb{Z}$). No division is used; this matters because in $\mathbb{Z}/m\mathbb{Z}$ division is not always possible (for instance, 2 has no inverse in $\mathbb{Z}/4\mathbb{Z}$).

### 2.5 Squaring modulo 2

**Lemma 2.1** (Frobenius identity)**.** *Let $p$ be a prime. For every power series $U$ with coefficients in $\mathbb{F}_p$, we have $U(x)^p = U(x^p)$.*

*Proof.* In $\mathbb{F}_p$, the binomial theorem simplifies to $(a+b)^p = a^p + b^p$, because every binomial coefficient $\binom{p}{i}$ with $0 < i < p$ is divisible by $p$. Also $c^p = c$ for every $c \in \mathbb{F}_p$ (Fermat's little theorem). So $\left(\sum_n u_n x^n\right)^p = \sum_n u_n^p x^{pn} = \sum_n u_n x^{pn} = U(x^p)$. Only finitely many terms matter for any given coefficient, so this is a polynomial identity. □

For $p = 2$ this says that squaring a series modulo 2 doubles every exponent: for instance, $(x + x^3)^2 \equiv x^2 + x^6 \pmod{2}$. In particular, the square of any series has, modulo 2, no terms of odd degree.

## 3 Results

We take formula (1) of the entry as the definition:

$$A(x) = x + A\big(M_1(x)\big)^2, \qquad M_1(x) = \frac{x}{1-x}. \tag{2}$$

**The nested expression.** To give the title a precise meaning, cut the nesting off after $d$ levels. That is, for $d \geq 1$ define series $H_d^{(d)}, H_{d-1}^{(d)}, \ldots, H_1^{(d)}$ from the inside out by

$$H_d^{(d)} = M_d, \qquad H_j^{(d)} = M_j + \big(H_{j+1}^{(d)}\big)^2 \quad (j = d-1, \ldots, 1),$$

and put $A^{[d]} = x + \left(H_1^{(d)}\right)^2$. For example,

$$A^{[3]} = x + \Big(M_1 + \big(M_2 + M_3^2\big)^2\Big)^2.$$

These are power series, obtained by finitely many additions, multiplications and divisions by $1 - cx$.

**Theorem 3.1** (What the nested square means)**.** *(a) Equation* (2) *has exactly one normalized solution $A$ with integer coefficients (and, more generally, exactly one with coefficients in any commutative ring).*

*(b) For $j \geq 0$ put $G_j = A(M_j(x))$. Then $G_0 = A$ and*

$$G_j = M_j + G_{j+1}^2 \qquad \text{for every } j \geq 0.$$

*This is formula (5) of the entry.*

*(c) For every $d \geq 1$, the series $A^{[d]}$ and $A$ have the same coefficients up to degree $d+1$: $A - A^{[d]} = O(x^{d+2})$.*

Part (c) says that the coefficient of $x^n$ in $A^{[d]}$ no longer changes once $d \geq n-1$. The infinitely nested expression of the title stands for the series made of these stable coefficients, and part (c) says that this series is $A$. So the definition in the title and formula (1) agree, and the sequence A392210 is $a(n) = [x^n]\, A$.

**Theorem 3.2.** *For every $n > 1$, the coefficient $a(2n-1)$ is even.*

# 4 Proofs

**The idea in brief.** The right-hand side of (2) contains $A$ only inside a square, and squaring "pushes" differences to higher degree; this gives uniqueness. Substituting $M_j$ into (2) and using $M_1(M_j) = M_{j+1}$ gives the nested form, and each level of nesting fixes one more coefficient. Finally, modulo 2 the square of a series has only even powers of $x$, so the only odd power of $x$ in $A$ is $x$ itself.

**Lemma 4.1** (An equation with a square)**.** *Let $T$ and $P$ be normalized series with coefficients in a commutative ring $R$. There is exactly one normalized series $G$ with coefficients in $R$ such that*

$$G(x) = T(x) + \big(G(P(x))\big)^2.$$

*If $T$ and $P$ have integer coefficients, then so does $G$, and reducing $G$ modulo $m$ gives the solution of the same equation over $\mathbb{Z}/m\mathbb{Z}$ (with $T$ and $P$ reduced modulo $m$).*

*Proof.* The point is that the coefficient of $x^n$ on the right-hand side depends only on the coefficients of $G$ below degree $n$. To see this, let $G$ and $H$ be two series with constant term 0 that agree below degree $r$, so $G - H = O(x^r)$. Since $P$ starts with $x$, the power $P^k$ has no terms below $x^k$, and so $G(P) - H(P) = O(x^r)$ as well. Moreover $G(P) + H(P)$ has constant term 0. Hence

$$G(P)^2 - H(P)^2 = \big(G(P) - H(P)\big)\big(G(P) + H(P)\big) = O(x^{r+1}).$$

In other words, changing $G$ in degree $r$ and above changes the right-hand side only in degree $r+1$ and above.

So the equation fixes the coefficients of $G$ one at a time. The constant term is 0, the coefficient of $x$ is the coefficient of $x$ in $T$, which is 1, and for $n \geq 2$ the coefficient of $x^n$ in $G$ must equal the coefficient of $x^n$ in $T + G(P)^2$, which has already been computed from the earlier coefficients. Only additions and multiplications are used, so this works in every commutative ring, and gives integer coefficients when $T$ and $P$ have them. Finally, reducing the integer solution modulo $m$ gives a normalized series that satisfies the reduced equation (reduction is compatible with all the operations involved), and by uniqueness it is *the* solution over $\mathbb{Z}/m\mathbb{Z}$. □

*Proof of Theorem 3.1.* (a) This is Lemma 4.1 with $T = x$ and $P = M_1$.

(b) Since $M_0(x) = x$, we have $G_0 = A$. Substitute $M_j(x)$ for $x$ in (2). Substitution can be done in steps, and substituting into a square is the same as squaring the substituted series, so

$$A(M_j) = M_j + \Big(A(M_1(M_j))\Big)^2 = M_j + A(M_{j+1})^2,$$

using $M_1(M_j) = M_{j+1}$ from (1). This is $G_j = M_j + G_{j+1}^2$.

(c) Fix $d \geq 1$, and set $H_{d+1} = 0$, $H_j = H_j^{(d)}$ for $1 \leq j \leq d$, and $H_0 = A^{[d]}$. Then $H_j = M_j + H_{j+1}^2$ for $0 \leq j \leq d$, the same rule as in part (b). We show by downward induction on $j$ that

$$G_j - H_j = O(x^{\,d+2-j}) \qquad (0 \leq j \leq d+1).$$

For $j = d+1$ this says $G_{d+1} = O(x)$, which is true because $G_{d+1}$ has constant term 0. If the claim holds for $j+1$, then

$$G_j - H_j = G_{j+1}^2 - H_{j+1}^2 = (G_{j+1} - H_{j+1})(G_{j+1} + H_{j+1}) = O(x^{\,d+1-j}) \cdot O(x),$$

because $G_{j+1}$ and $H_{j+1}$ both have constant term 0. This is $O(x^{d+2-j})$, as claimed. For $j = 0$ we get $A - A^{[d]} = O(x^{d+2})$. □

*Proof of Theorem 3.2.* Reduce (2) modulo 2; this is allowed because reduction is compatible with all the operations involved. By Lemma 2.1, for every series $U$ with coefficients in $\mathbb{F}_2$ we have $U(x)^2 = U(x^2)$, which contains only even powers of $x$. Applying this to $U = A(M_1(x))$, we see that, modulo 2, the right-hand side $x + A(M_1)^2$ has no odd powers of $x$ except $x$ itself. Hence the coefficient of $x^{2n-1}$ in $A$ is even whenever $2n - 1 \geq 3$, that is, whenever $n > 1$. □

## 5 Checking the results

A mathematical proof can contain mistakes, so the results were checked in two independent ways.

**Numerical check.** We computed $A$ by repeating the substitution $A \mapsto x + A(M_1)^2$ with exact integer arithmetic, starting from $A = x$. This reproduces all 24 terms printed in the OEIS entry. We also evaluated the nested expression of the title, cut off after 202 levels, and found the same coefficients below degree 200, as Theorem 3.1(c) predicts. All coefficients of odd degree $2n - 1$ with $3 \leq 2n - 1 < 200$ are even. The computation was done with the PARI/GP computer algebra system, using code that was written separately from the formal proofs (file `verification/independent_checks.gp` in the accompanying repository [7]). A computation can only test finitely many cases, so it is a check on the definitions and on the indexing, not a proof.

**Formal proof in Lean.** Lean [5] is a *proof assistant*: a program in which mathematical statements and their proofs are written in a precise formal language, and every step is checked mechanically down to the basic rules of logic. All results of this paper are proved in Lean 4 (version 4.19.0) on top of Mathlib [6], Lean's library of formalized mathematics. The statements are collected in the file `lean/Entries/A392210.lean` of the repository, in the namespace `OEIS.A392210`. There $x/(1-x)$ is called `geometric`, and the proofs are in the module `lean/Families/NestedBinomial.lean`. Parts (b) and (c) of Theorem 3.1, which connect formula (1) with the nested expression of the title, are proved in this paper but are not part of the Lean formalization. The table lists where each result of this paper is proved.

| Statement in this paper | Lean theorem(s) |
|---|---|
| Theorem 3.1(a) for integer coefficients | `third_existence` |
| $A$ is normalized and satisfies (2); and $M_1(x) \cdot (1-x) = x$ | `third_definition` |
| Theorem 3.2, for $A$ and for any normalized integer solution of (2) | `third_evenness`, `third_arbitrary_source` |

The Lean project was rebuilt from source for this paper, and Lean was asked to list the basic assumptions (axioms) that each of these theorems ultimately depends on. Every one depends only on the three standard axioms of Lean's logic (`propext`, `Classical.choice`, `Quot.sound`); no step is left unproved. A formal proof certifies exactly the statement that Lean checks. Whether that statement says what the OEIS entry means is a separate question, so the Lean definitions were also compared, side by side, with the definition in the OEIS entry (Section 1.1).

# 6 Discussion

**Scope.** Theorem 3.2 proves the observation of the entry for every $n$. It is an elementary fact, and the point of this paper is to give a complete and checked account of it, including the meaning of the nested expression. The theorem says nothing about the terms of even index, nor about remainders modulo 4 or other numbers.

**The even-indexed terms.** The same idea gives a rule for the parity of the remaining terms. Apply Lemma 2.1 to $U = A(M_1(x))$ more carefully: modulo 2, $U(x)^2 = U(x^2) = A\big(M_1(x^2)\big)$. So

$$A(x) \equiv x + A\Big(\frac{x^2}{1-x^2}\Big) \pmod 2.$$

Comparing the coefficients of $x^{2m}$ on both sides gives, for every $m \geq 1$,

$$a(2m) \equiv \sum_{k=1}^{m} \binom{m-1}{k-1} a(k) \pmod 2.$$

So the parity of $a(2m)$ is determined by the parities of $a(1), \ldots, a(m)$. (This rule is not part of the Lean formalization; we checked it numerically for all $2m < 130$.) Unlike the odd-indexed terms, the even-indexed terms show no simple pattern in our data.

**Other equations of the same kind.** The entry lists further identities of the same shape, for example $A(x/g(x)) = x/g(x) + A\big(x/(g(x)-x)\big)^2$ for every series $g$ with constant term 1. All of them follow from formula (1) by substitution, in the same way as part (b) of Theorem 3.1.

## About this paper

**How the work was done.** This paper is one of 26 papers on conjectures from the OEIS that came out of an autonomous AI research effort directed by Dev D. Goyal [7]. The roles were as follows.

- *Dev D. Goyal* started the project, chose which problems to pursue, asked for proofs, formal verification and revisions, and decided what to publish. Dev D. Goyal did not supply proof strategies or intermediate arguments.
- *ChatGPT* (OpenAI), using the GPT-6 model with Pro reasoning, worked autonomously from those prompts. It searched the OEIS for suitable conjectures, found the proofs, wrote the Lean 4 formalization, and wrote a first draft of this paper.
- *Claude Opus 5.5*, used with Claude Code, checked the work independently and then rewrote this paper. It rebuilt the Lean project from source and confirmed that the theorems listed in Section 5 rely only on Lean's standard axioms; compared the Lean definitions with the OEIS definition; recomputed the sequence with separately written PARI/GP code, compared it with the terms published in the OEIS, and tested every claim of the paper on many terms; and read every step of the proofs. It then rewrote the text to make it self-contained and readable. The theorems and the proof strategies are ChatGPT's. In rewriting, Claude Opus 5.5 expanded the explanations and, in places, wrote out an argument in more detail or by a more elementary route. The Discussion section was added by Claude Opus 5.5; the further observations in it were checked by computer and are not part of the Lean formalization.

Dev D. Goyal has verified the results and is responsible for this paper.

**Review status.** This paper has not been peer reviewed, and it has not been submitted to a journal or to the OEIS.

**Priority.** An OEIS entry that labels a statement as a conjecture shows only that its contributor did not know a proof; it does not show that no proof existed. During the research only limited searches for earlier work were made. No claim of priority is made.

**Corrections.** Please report any error in the issue tracker of the repository [7], where the Lean sources and the numerical checks are also available.

## Acknowledgements

The sequence, its formulas and the observation are due to Paul D. Hanna [1].

## References

[1] P. D. Hanna, entry A392210 (G.f.: $x + \mathrm{sq}(x/(1-x) + \mathrm{sq}(x/(1-2x) + \cdots))$, an infinitely nested square), *The On-Line Encyclopedia of Integer Sequences*, February 2026. `https://oeis.org/A392210`. Text as of October 2026.

[2] N. J. A. Sloane and The OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*. `https://oeis.org`.

[3] R. P. Stanley, *Enumerative Combinatorics*, Volume 2, Cambridge University Press, 1999. (Chapter 5 treats formal power series, composition and compositional inverses.)

[4] I. Niven, Formal power series, *American Mathematical Monthly* 76 (1969), 871–889.

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